Kanay Gupta

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EDUCATION

University of New South Wales (UNSW) February 2023 – Expected December 2026
Bachelor of Advanced Computer Science (Honours)

- WAM: 87 (High Distinction). Faculty of Engineering Dean's Honours List (2023).

EXPERIENCE

ResMed May 2026 – Present
Student Intern – Software Engineer, Digital Platform Engineering
Tech Stack: Java (Spring), TypeScript/Angular, Python, AWS (EKS, ECS), Kubernetes, Terraform, GitHub Actions

- Delivered 40+ features, defect fixes, and migrations on a cloud device simulation platform that about 18 product and V&V teams use to test ResMed's connected CPAP and ventilator devices, across a Java REST API, an Angular portal, data generation engines, and a Python MCP server.
- Built simulation support for the AirSense 11 Pro launch, adding settings acknowledgement, schema-version-aware IoT endpoint routing, a review-and-retry flow for failed data sends, filter presets, and Care Check-In data, and fixed Protobuf encoding defects in its new JSON-schema data engine.
- Implemented Breath-by-Breath data delivery controls, initialising control flags at device creation and simulating device acknowledgement of control and over-the-air update messages, then deployed the release to production.
- Migrated the web portal and API from ECS to the company's shared Kubernetes (EKS) platform and moved 10+ services, including an AWS China deployment, from NGINX Ingress to Envoy Gateway, updating Helm values, Terraform load balancer rules, and GitHub Actions pipelines.
- Extended an MCP server that lets testers set up simulations in natural language, adding compliance-device assignment and per-environment default users, and fixed intermittent connection resets and spurious 400 API responses.

Satire Explanation with Language Models (Honours Thesis) February 2026 – Present
Undergraduate Honours Researcher (UNSW Computer Science and Engineering)
Tech Stack: Python, PyTorch, HuggingFace, OpenRouter, Gradio, Llama-3.1, Qwen2.5, Mistral, Katana HPC

- First-author paper accepted at ALTA 2026: *Unreliable Narrator Framing as a Cost-effective Mechanism for Satire Explanation of Australian and American Satirical News*, with David Y. Liu and Dr. Aditya Joshi.
- Built a 1,000-article benchmark of Australian (The Betoota Advocate) and American (The Onion) satirical news to test whether Large Language Models (LLMs) can explain satire that depends on recent, local context.
- Designed an "unreliable narrator" prompting framework grounded in narrative theory that improved judged explanation quality in 16 of 17 model and dataset comparisons (+0.27 to +0.90 on a 5-point scale, all significant), making 7–8B open-weight models a low-cost alternative to frontier models.
- Built an LLM-as-judge pipeline scoring six-dimension rubrics across 50+ generation runs on four open-weight and two frontier models, validated against three human annotators (judge correlation 0.44–0.76).
- Ran control experiments separating the narrative framing from prompt structure and answer length, and deployed a live Gradio web demo of the full pipeline with per-session and spending limits.

PROJECTS

LedgerMind – AI-Powered Tax Compliance App September 2025 – November 2025
Project Manager and Engineer (Capstone Project for IntelliServe)
Tech Stack: Python, React Native, PostgreSQL, Docker, OpenAI API, RAG, Jira, Git

- Led a 6-person agile team through weekly sprints and stakeholder meetings to deliver an AI tax app MVP.
- Built an LLM pipeline extracting vendor, date, and amount from receipt images, voice logs, and email invoices.
- Designed a RAG system validating expenses against live Australian Taxation Office (ATO) feeds.

AI-Powered Job Ad Extractor – LLM Comparative Analysis March 2025 – April 2025
AI Engineer (Industry Project for SEEK)
Tech Stack: Python, PyTorch, HuggingFace, Scikit-learn, Pandas, Gemini API

- Benchmarked three approaches to numerical extraction from job ads, then fine-tuned an open-weight Pythia-160m model with custom prediction heads, cutting Mean Absolute Error by over 95% against a Scikit-learn baseline.
- Annotated a 50+ sample evaluation benchmark with 94% inter-annotator agreement.

Ray Tracing Engine from Scratch in Rust – GitHub July 2025
Tech Stack: Rust, Multithreading

- Wrote a multithreaded path tracing engine from scratch without external rendering crates, implementing physically based materials (metals, dielectrics), texture mapping, and a configurable depth-of-field camera.
- Achieved a 7.16x rendering speedup on a 16-core CPU by parallelising the workload using Rust's native concurrency primitives (Arc, Mutex).

SKILLS

- **Languages:** Python, Java, TypeScript/JavaScript, Rust, C#, SQL, HTML/CSS
- **Frameworks and Libraries:** PyTorch, HuggingFace, Scikit-learn, Pandas, Spring Boot, Angular, React.js, Next.js, Node.js
- **Tools and Platforms:** AWS (EKS, ECS, DynamoDB, SQS), Kubernetes, Helm, Terraform, Docker, GitHub Actions, Git, PostgreSQL, Linux, Jira